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Short Note

Co-deposition of TiO_2/CdS films electrode for photo-electrochemical cells

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Abstract

A densely packed TiO_2 thin film onto an indium doped–tin oxide (ITO) substrate was synthesized at room temperature by chemical deposition and a CdS thin film was deposited onto the pre-deposited TiO_2 film by a doctor blade route (powder of CdS was obtained from chemical deposition). TiO_2/CdS film was annealed at 300 °C for 1 h in air for crystallinity improvement. The first grown TiO_2 film was nanocrystalline, whereas the CdS film was polycrystalline as evidenced by X-ray diffraction (XRD) and selected area electron diffraction (SAED). Scanning electron microscopy (SEM) images show formation of mono-dispersed CdS spherical grains onto compact, densely packed spherical nanocrystalline grains of TiO_2. The TiO_2/CdS bilayer film was used in a photo-electrochemical cell as a working electrode, and a platinum electrode as a counter electrode (0.1 M lithium iodide electrolyte) under 80 mW/cm^2 light illumination intensity.
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1. Introduction

Nanocrystalline titanium dioxide thin films have attracted both industrial and academic attention for potential applications as the gate dielectric in metal oxide semiconductor field effect transistors (Peercy, 2000), as a surface for solar energy conversion (Bach et al., 1998), as high efficiency photocatalysts (Dohshi et al., 2003), and as gas sensors (Skubal et al., 2002) because of their well-known properties, including high refractive index, and transmittance in visible region. In particular, for the purification of air and water from contaminants catalytic approach plays a vital role, which is based on the feature that the TiO_2, including some other semiconductors, is able to photo-catalyze many organic substances under UV irradiation (Fujishima et al., 2000). Apart from the high photoactivity, TiO_2 is biologically and chemically inert, abundantly available, and cheap. Due to donor-like oxygen vacancies TiO_2 is an n-type semiconductor.

Polycrystalline CdS thin films have received considerable attention during the recent years because of their proven and potential applications in the photoconductor as well as photovoltaic devices. CdS films have been prepared by various methods such as vacuum deposition, sintering, spray pyrolysis, chemical deposition etc. (Vigil et al., 1998; Nanda et al., 1998; Ristova and Ristov, 1998). Many researchers have reported (Prabahar and Dhanam, 2005; Cui and Xi, 1996) synthesis of CdS films from aqueous and non-aqueous baths. It is well known that the couplings

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of two or more semiconductors decrease electron/hole recombination with improved catalytic activity of the micro-heterojunction (Lokhande et al., 2005). Oxidation of aniline on CdS by solar photocatalysis is recently reported (Karunkarn and Senthilvelan, 2005). When CdS is used in combination with TiO_2, charge transfer can be achieved through selective transfer of conduction band electrons from CdS to TiO_2 nanoparticles, the transfer of holes being inhibited for energetic reasons (Tambwekar et al., 1999; Borgarello et al., 1985).

CdS films are useful as (buffer) layers for certain devices, but if deposited at low temperature, good crystallinity is not expected. In literature, CdS thin films on CdTe polymer based flexible substrate for 11.3% photo-conversion efficiency (Mathew et al., 2004) were used. In this paper, we report three-step process for TiO_2/CdS coupled thin film synthesis. The first step involves TiO_2 thin film synthesis on ITO substrate by hydrolysis of titanium tri-chloride solution at room temperature from aqueous acidic bath. In the second step, CdS powder was prepared by chemical bath process, and in the third step, CdS paste was pressed using doctor blade route in which CdS paste formed with suitable surfactant such as methanol is pasted on ITO/TiO_2 film by carefully fixing spacer on both the sides. Paste of CdS was pressed with other slide in-between the spacers to form TiO_2/CdS electrode. TiO_2/CdS bilayered film so formed was characterized for its structural, surface morphological and photo-electrochemical properties by means of various techniques.

2. Chemical reactions and experimental procedure

Slow hydrolysis of titanium tri-chloride in a water solution was used for the deposition of titanium dioxide thin films on ITO substrate. This reaction consumes H_2O and generates OH^- species with increase in pH near the substrate. The energy required to hydrolyze titanium tri-chloride precursor is provided by the liquid in the bath, either by thermal collisions or by a chemical reaction between the liquid medium and the precursors, or by a combination of these two mechanisms. Reaction of titanium with oxygen is the rate determining step during the titanium dioxide formation in which titanium hydroxide ions is the only reactive species (Lokhande et al., 2004) as

$$Ti^{3+} + 3OH^- \rightarrow Ti(OH)_3 \quad \text{and}$$
$$Ti(OH)_3 + OH^- \rightarrow TiO_2 + 2H_2O \tag{1}$$

pH of the resultant solution is found to play vital role during film deposition and the surface morphology of the product which was maintained about 3.5 with the drop wise addition of ammonium hydroxide solution at room temperature. Film of titanium dioxide formation starts after the time period of 15 min and completed within 5 h at room temperature. Dipping for longer time period of substrates in deposition bath were results peeling off already deposited film due to dissolution mechanism.

At this stage, we attempted to deposit CdS film directly onto TiO_2 many times under different deposition conditions. However, during each time of CdS deposition, TiO_2 film was found to be removed from ITO substrate, which made us find another route for CdS deposition to make TiO_2/CdS bilayer film i.e., doctor blade.

For CdS powder synthesis, chemical bath deposition of 0.1 M cadmium chloride and thiourea (ammonia complex) was carried out at 60 °C for 1 h, since rate of release of Cd^{2+} ions from the ammonia complex affects rate of film deposition (for details see Mane and Lokhande, 1997). Paste was prepared by constantly stirring CdS powder in polyethylene glycol (PEG) [$-O-(CH_2-CH_2-O)_n$ (M.W. 10,000)] polymer binder solution (0.5 g CdS powder in 4 mL deionized water along with 20 wt% PEG) for 2 h. Carefully fixing the self adhesive tape on either sides of substrate, CdS paste was applied using a doctor blade to form a relatively thick film onto pre-deposited TiO_2 film. TiO_2/CdS bilayered film so formed was annealed in air at 300 °C for 30 min and further used for characterization with techniques including XRD, SEM, SAED. Weight difference method was preferred for thickness measurement for both TiO_2 and CdS films. The X-ray diffraction patterns were obtained by using X-ray diffractometer (RINT/PMAX 2500, Rigaku, Japan, Cu-Kα radiation of wavelength 1.5428 Å) and SEM images were taken from FE-SEM, (SM-6340F, Jeol, Japan) operated at 15 kV. For SAED, film of TiO_2 was scratched from substrate and mounted on copper grid by means of micro-holder. Photo-activity was studied by forming photo-electrochemical cell with TiO_2/CdS as a photo-electrode and platinum as a counter electrode. The electrolyte used was 0.1 M lithium iodide. A xenon lamp (450 W, Thermo Oriel Co., USA) with illumination intensity of 80 mW/cm^2 was used as a light source.

3. Results and discussion

TiO_2 and CdS film thicknesses measured by weight difference method were 180 nm, and 560 nm, respectively. Fig. 1 shows crystalline structure of CdS film, deposited by doctor blade onto pre-deposited TiO_2 film as suggested by well-defined and relatively high intensity CdS peaks [JCPDF Nos. 47-1179, 43-0985, and 21-0828]. Unknown peak at 61.92° was notified. In X-ray diffraction pattern (not shown separately) of TiO_2, no practical difference was observed corresponding to ITO substrate, which indicates it's amorphous and/or nanocrystalline nature. SAED pattern of TiO_2 shows a set of rings instead of spots due to the random orientation of crystallites, corresponding to diffraction from different planes of nanocrystallites. The location of planes corresponding to (1 1 0), (1 1 2), (1 1 1), (2 1 1), (0 5 2), and (3 0 1) are in good agreement with the [JCPDS File No. 21-1276] reference diagrams for the corresponding bulk phases with lattice constants $a = 4.593$ Å, and $c = 2.959$ Å, respectively. The rings corresponded to

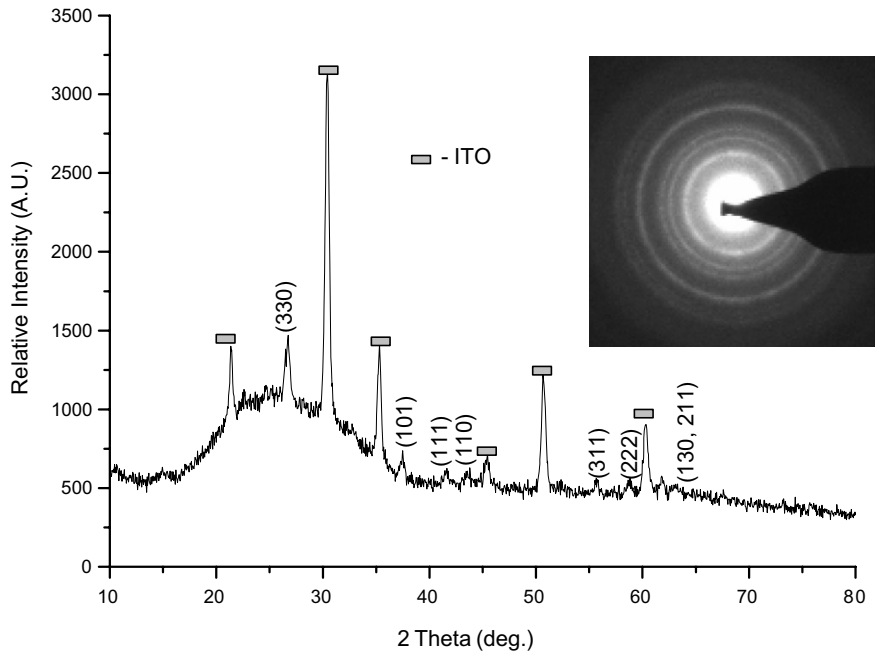

Fig. 1. XRD pattern of nanocrystalline TiO_2/CdS film on ITO substrate (inset shows selected area electron diffraction pattern of TiO_2 film, since more noise peaks in its XRD were detected).

specific rutile lattice planes of the nanocrystallites of TiO_2. Clear, sharp, and distinct circular rings supported for nanocrystalline nature (Seo et al., 2003). Films with nearly diffused or fuzzy rings are corresponding to significant amorphous structure (Han et al., 2004).

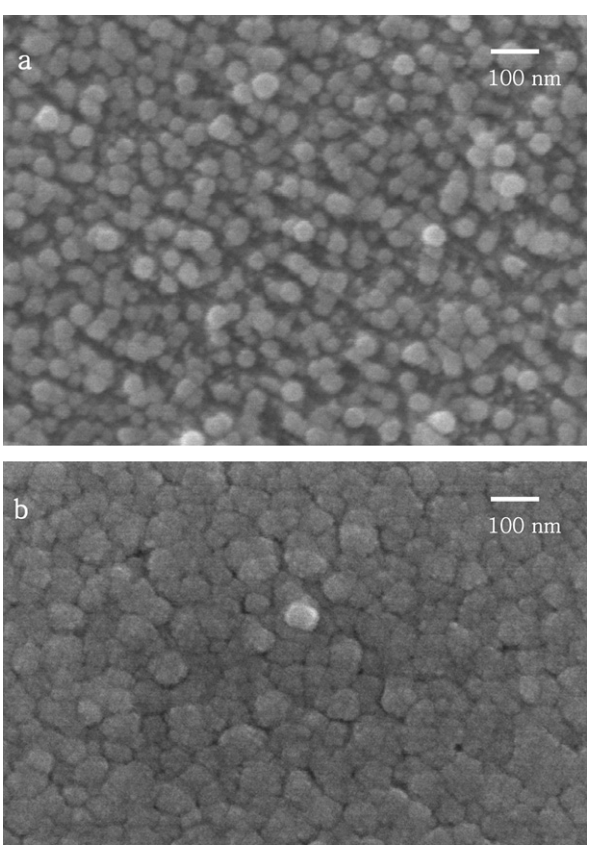

Fig. 2. SEM images of TiO_2 (a), and TiO_2/CdS bilayer (b) thin films.

SEM images of TiO_2 and TiO_2/CdS films are shown separately in Fig. 2(a,b). The TiO_2 film is composed of closely packed nanometer sized spherical grains, whereas TiO_2/CdS bilayered film consists of relatively mono-dispersed spherical grains of CdS on TiO_2. Films surface after CdS film formation reveals no cracks or void spaces confirming an effective role of binder. However, surface roughness seems to be more with little aggregation of grains.

In order to demonstrate the potential use of chemically and doctor blade deposited TiO_2/CdS film, photo-activity of TiO_2/CdS film in 0.1 M lithium iodide electrolyte was studied. Fig. 3 shows *I–V* characteristics of TiO_2/CdS film deposited on ITO substrate. While *I–V* measurement, 0.28 square centimeter area of TiO_2/CdS film was exposed to light. Fig. 3 shows dark (○) and light illumination (□)

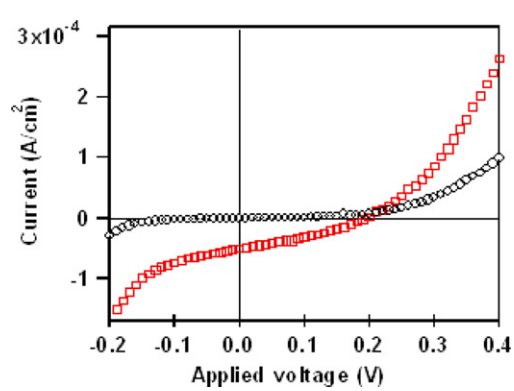

Fig. 3. *I–V* curves of TiO_2/CdS bilayer film with (□) and without light illumination (○).

curves for TiO_2/CdS. No photo-corrosion was observed even after scanning the film for ten times. Here we believe that CdS with PEG polymer excite electrons and transport into conduction band of TiO_2 (Manorama et al., 2002). Observed low fill factor (37%) is attributed to amorphous/crystalline interface developing potential barrier for free motion of excited electrons. Surface morphology difference between TiO_2 and CdS nanoparticles suggested need of further efforts to prepare TiO_2/CdS bilayer film for better performance.

4. Conclusions

TiO_2/CdS aqueous system with three-deposition steps for fabrication of TiO_2/CdS bilayer thin film on ITO substrate at low temperature has been successfully developed. TiO_2 film was nanocrystalline nature and composed of densely packed spherical nanometer sized grains, whereas CdS film consists of polycrystalline structure with mono-dispersed spherical grains. The TiO_2/CdS bilayer electrode so formed was efficient in photo-electrochemical cells as a working electrode.

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